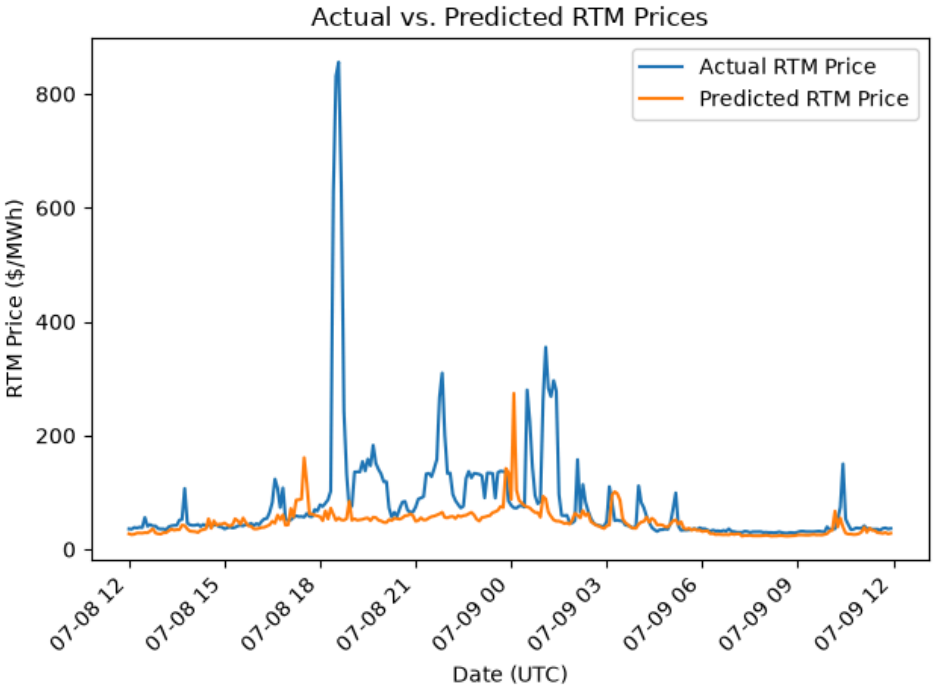
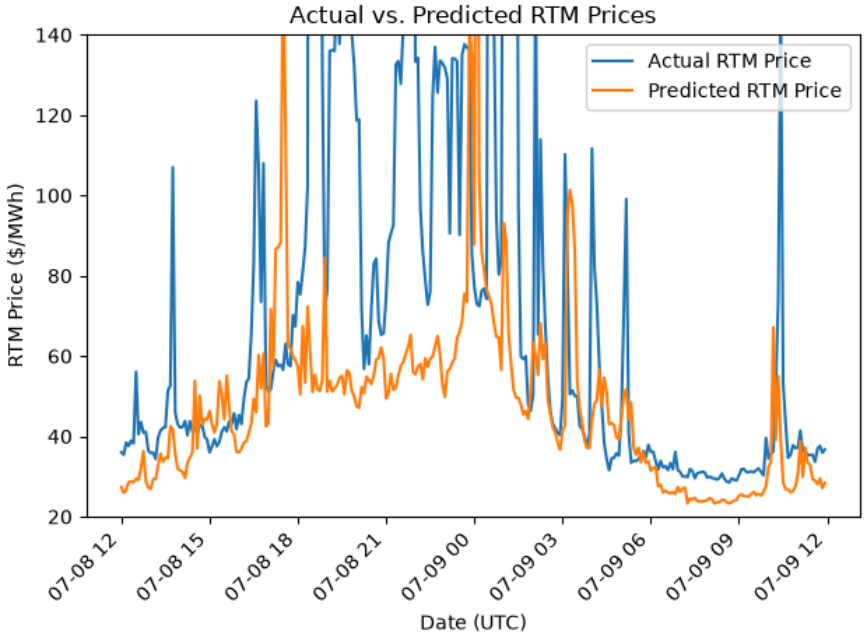
Our Approach To Outliers:

Our target variable, LMP Price, has several outliers, or prices that are much higher than all of the other prices. These are the result of unexpected conditions that occur (somewhat) regularly in the market, typically sudden loss of operating equipment which makes it difficult to get the power where it needs to be which drives up the prices to extreme values.

These extreme values create a lot of noise for predictions, as seen below:

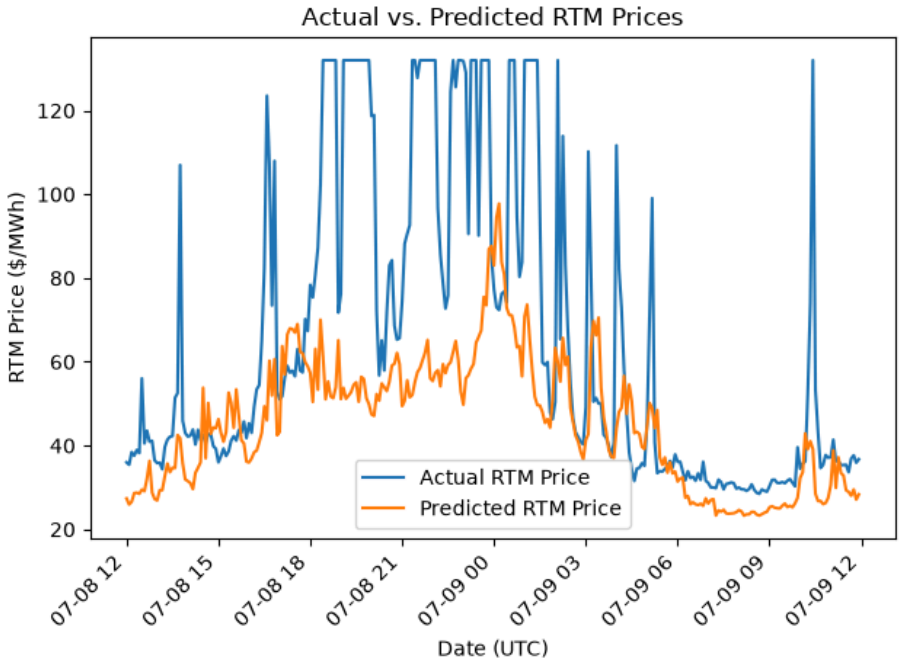


Zoomed in version:



We are exploring two options to deal with this. The first option is to do nothing, while these outliers hurt our prediction RMSE's and other metrics, at the end of the day, we don't really care about RMSE, we only care about how well the optimization algorithms (both the RL optimization and the LP optimization) works (how much profit it makes), so one option is to leave these extreme values in, so that the algorithm can get a good idea of the potential extremes for a given day.

An alternative way to handle the outliers is to set a cutoff threshold, because once energy prices go above a certain threshold, the magnitude of the price more or less becomes irrelevant. The idea of this method is that when training the model, it will have less noise to deal with, allowing the model to better follow the daily trends, without being as affected by the extreme spikes. When predicting on the same data predictions look like this:



It is impossible to tell at this stage which method will work better as input for the optimization algorithm, so the team will plan to move forward with both options in order to evaluate both on the optimization algorithm in the future. An important consideration to make is that we can see a visible decrease in predictions after the removal of outliers. This is the expected outcome because there are lower rolling averages, which is what the baseline model uses for predictions. But we can expect a similar pattern for other models, since there is no reward for predicting anything over 132 and less penalty for underpredicting. But when we get to the real data there will be no outlier removal, and the lower predictions will again have higher error and will actually be less accurate than without outlier removal.

Baseline Model

Model: Moving Averages (Baseline)

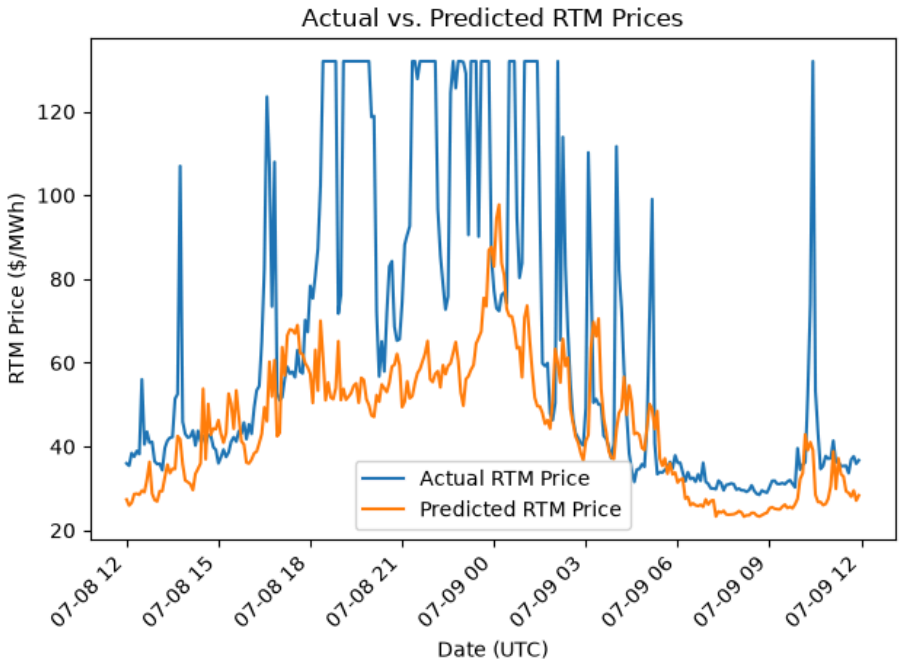
Hyperparameters:

- Lookback- Values: 288 steps (1 day), 864 steps (3 days), 2016 steps (7 days), and 4032 steps (14 days)

Tuning Summary:
Outliers capped at 132

Lookback Steps	MAE ($)	RMSE ($)
288	17.10	27.28
864	16.18	24.77
2016	15.24	23.41
4032	15.66	23.97

Graph:



Moving Averages - 2016 Lookback Steps Model

Results/Summary:

This model, for the most part, predicts where the spikes in price are however struggles with accurately predicting the magnitude of the price spikes. In general though the predictions follow the general trend of the data, at least loosely. The final R^2 value for this model was 0.220.

Sarimax Model

Model: SARIMAX

Hyperparameters:

- p (Autoregressive order) - Values: 1 and 2
- d (Moving Average order) - Values: 0, 1, and 2
- lookback - Values: 2016 and 4032 (1 and 2 weeks)
- fourier_k - Values: 2, 4, and 6
- exog_cols - Combinations of: ["lmp_dam", "load_forecast", "north_wind", "central_wind", "south_wind", "north_solar", "central_solar", "south_solar"]

Tuning Summary:

outliers capped at 1000:

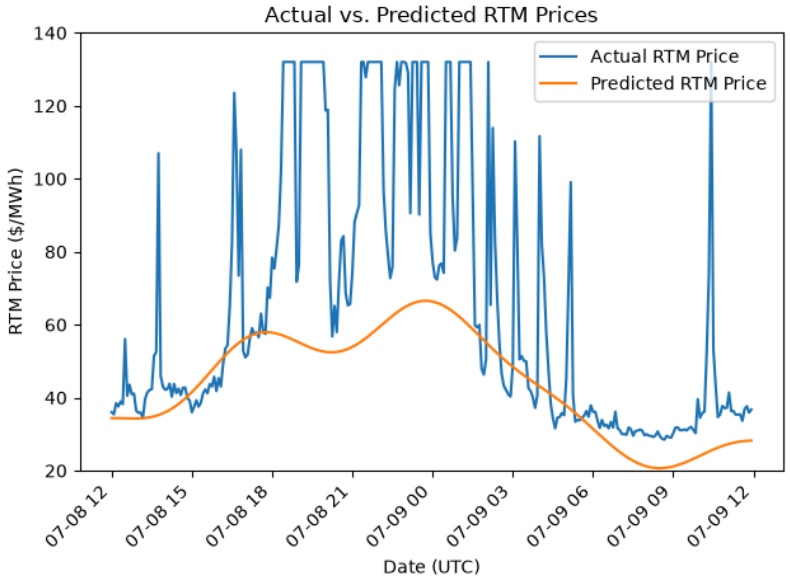
id	order [p, 0, d]	lookback	fourier_k	exog_cols	val_mae	val_rmse
1	[1, 0, 1]	2016	4	None	27.029313	68.948522
0	[1, 0, 1]	2016	2	None	27.043550	70.219907
5	[1, 0, 1]	2016	4	load_forecast, miso_wind, miso_solar	62.115922	94.596735
8	[1, 0, 1]	2016	4	lmp_dam, load_forecast, miso_wind, miso_solar	61.483737	97.448252
14	[1, 0, 1]	2016	4	lmp_dam, load_forecast, miso_wind, miso_solar	61.779842	102.800755

outliers capped at 132:

id	order	fourier_k	lookback	exog_cols	val_mae	val_rmse
1	[1, 0, 1]	4	2016	None	17.58	27.19
0	[1, 0, 0]	2	2016	None	18.76	28.70
8	[1, 0, 1]	4	2016	lmp_dam, load_forecast, miso_wind, miso_solar	57.33	73.45

14	[1, 0, 1]	4	2016	lmp_dam, load_forecast, north_wind, central_wind, south_wind, north_solar, central_solar, south_solar	56.42	76.56
5	[1, 0, 1]	4	2016	load_forecast, miso_wind, miso_solar	59.24	80.82
2	[1, 0, 1]	4	2016	lmp_dam	314.20	1197.64
4	[1, 0, 1]	6	4032	lmp_dam	401.89	1436.35

Graph:



SARIMAX - Best Hyperparameter Combination

Results/Summary:

While the RMSE values were somewhat promising, the SARIMA model struggled to fit the rapid changes in prices, and any increases in model complexity exponentially increased the run time of the model due to the number of predictions that the model needed to make. Ultimately we decided not to continue with this model because of the time it takes to run, while still not coming close to the baseline model (which runs almost instantaneously).

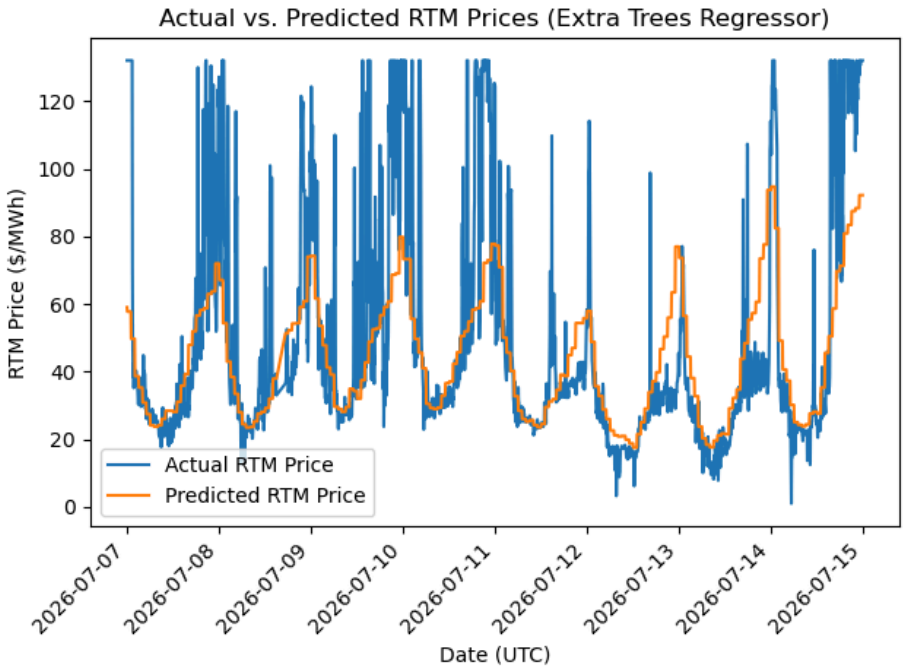
Non-Time series Models: Extra Trees Regressor

Hyperparameters:

- N_estimators: 800
- Max_features: 1.0
- Max_depth: 10
- Min_samples_split: 2
- Min_samples_leaf: 8
- Max_leaf_nodes: 100

Outliers capped at 132.

Graph:



Best Hyperparameter Combination for Extra Trees Regressor

Results/Summary:

	RMSE	**R2**
Validation	20.18	0.40

Testing	19.85	0.45

For the non-time series models, Extra trees regressor was showing the best early results so most focus went into this one. After tuning the hyperparameters, the RMSE improved by about 1. RandomizedSearchCV was used to find the best hyperparameters and is what gave the results shown above. For the hyperparameters listed above, those were the best options chosen from a list of 5 or more for each hyperparameter. As the Extra Trees Regressor has been tuned and improved, the prediction is fitting the graph much more nicely than it was several weeks ago. The outliers were imputed to 132 as that was what our group agreed upon as detailed previously.

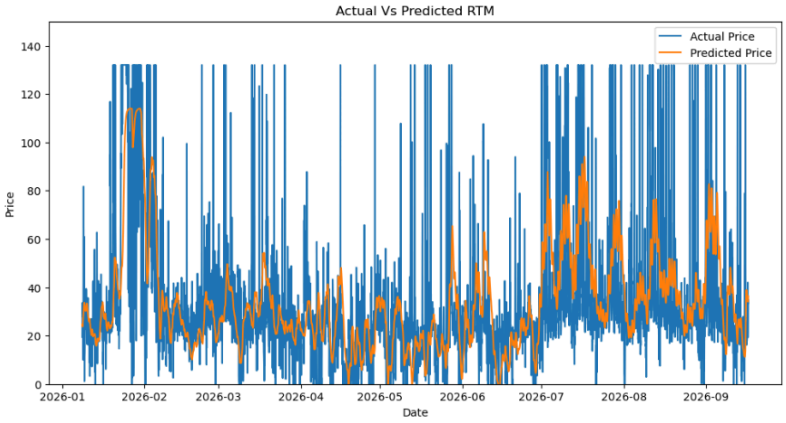
Long Short Term Memory Model

Hyperparameters:

- Hidden units: 64, 128
- Window size: 288 (1 day), 2016 (7 days)
- Learning rate: 0.0001, 0.001
- Dropout: 0.1
- Layers: 2

Tuning Summary:

Window Size	Hidden Units	Learning Rate	RMSE	R Squared
288	64	0.0001	24.17	0.2
288	128	0.0001	24.24	0.19
288	64	0.001	26.97	0
288	128	0.001	26.44	0.04
2016	64	0.0001	26.84	0.01



Results/Summary:

Though the RMSE values stay similar throughout different tuning parameters, the R squared values change drastically with different hyperparameters. The best R squared values observed were the results of a learning rate of 0.0001 and a window size of 24 hours. The change in the number of hidden units appeared to have only a slight impact. Decreasing the learning rate drastically decreased the R squared value to 0, indicating that there is little to no relationship

between predicted and actual prices. I chose the best performing model to run with a new window size of 7 days. This model was computationally expensive and drastically worse than its 24 hour window size counterpart. The best model only has a R squared value of 0.2. This could mean that the LSTM model is incredibly weak in predicting patterns in this data.

Future Work

RMSE has been what we are using to evaluate our models, but this is a flawed evaluation model because even with outlier removal, it still overestimates the error for outliers. We truly only care about the rank of the prices rather than the actual value. The value helps us determine how much money we gain or lose, but even if we gain 30 cents, that is still a decision the algorithm will choose just as much as an option where you gain $30 because in both cases the alternative is to gain nothing. As such, we are considering using R^2 as an evaluation metric instead of RMSE as it is more compatible with a rank based framework that we have in our environment.